

Due: September 11, 2026

This homework develops two complementary skills: (1) using R to explore and visualize data, and (2) implementing numerical optimization methods for maximum likelihood estimation. You should write clear, reproducible R code and briefly explain the important features of your numerical and graphical results.

1. Exploring the College Data

Use the dataset `College.csv`, available on Canvas. The data contain information about colleges and universities. Complete the following tasks.

- (a) **Import and inspect the data.** Use `read.csv()` to import the file into R as a data frame named `schools`. Then use `dim()`, `names()`, and `str()` to report the dimensions, variable names, and data types of the variables. Briefly identify which variables are quantitative and which are categorical.
- (b) **Set the university names as row names.** The first column contains the names of the universities. Use the following commands:

```
rownames(schools) <- schools[, 1]
schools <- schools[, -1]
View(schools)
```

Explain in one or two sentences why removing the university-name column from the data frame is useful after the names have been assigned as row names.

- (c) **Numerical and graphical exploration.** Use the data to address each of the following.
 - i. Use `summary()` to produce a numerical summary of the variables. Identify at least two variables for which the range appears substantially larger than the interquartile range, and briefly explain what this suggests about their distributions.
 - ii. Use `pairs()` to create a pairwise scatterplot matrix for the quantitative variables. To keep the figure readable, select an appropriate subset of quantitative variables if necessary. Identify two pairs of variables that appear to have a strong positive or negative association. State whether the relationship appears approximately linear.
 - iii. Create side-by-side boxplots of `Enroll`, separated by `Private`. Report which group has the larger median enrollment and comment on whether the variability appears similar between the two groups. Mention any potential outliers that you observe.

- iv. Create an ordered factor called `Faculty` according to

$$\text{Faculty} = \begin{cases} \text{"Moderate"}, & \text{PhD} < 75, \\ \text{"Good"}, & 75 \leq \text{PhD} < 85, \\ \text{"Great"}, & \text{PhD} \geq 85. \end{cases}$$

Use `factor(..., ordered=TRUE)` or an equivalent approach so that the ordering is explicitly recorded. Then create side-by-side boxplots of `Top10perc` by `Faculty`. Describe the trend in the median and spread across the three groups.

- v. Create histograms for four quantitative variables of your choice. For each histogram, specify either the number of breaks or the bin boundaries, and include an informative title and at least one useful axis label. Place all four histograms in a single figure using, for example, `par(mfrow = c(2,2))`.

For one of the four variables, make a second histogram using a different choice of breaks. In two or three sentences, explain how the choice of bin width affects the visual impression of the distribution.

- (d) **Reproducibility and interpretation.** At the end of Problem 1, include the R commands needed to reproduce your main numerical summaries and figures. Your submission should make it possible for another student to reproduce your results by running your code on the same data file.

2. Maximum Likelihood Estimation for Oil-Spill Data

There were 46 crude oil spills of at least 1000 barrels from tankers in U.S. waters during 1974–1999. The data file `oilspills.dat`, available on Canvas, contains the following information for each year i :

- N_i : the number of oil spills in year i ;
- b_{i1} : the estimated amount of oil shipped through U.S. waters as part of U.S. import/export operations, adjusted for spillage in international or foreign waters;
- b_{i2} : the amount of oil shipped through U.S. waters during domestic shipments.

Oil shipment amounts are measured in billions of barrels (Bbbl). The shipment volume represents the exposure to spill risk.

Suppose that, conditional on b_{i1} and b_{i2} ,

$$N_i \mid b_{i1}, b_{i2} \sim \text{Poisson}(\lambda_i), \quad \lambda_i = \exp(\alpha_1 b_{i1} + \alpha_2 b_{i2}),$$

where α_1 and α_2 are unknown parameters. Thus, α_1 and α_2 describe how the expected number of spills changes with the amount of oil shipped through the two types of operations.

You can read the data into R using

```
oil <- read.table(file = "oilspills.dat", header = TRUE)
```

The likelihood function is

$$L(\alpha_1, \alpha_2) = \prod_{i=1}^n \frac{\lambda_i^{N_i}}{N_i!} \exp(-\lambda_i), \quad \lambda_i = \exp(\alpha_1 b_{i1} + \alpha_2 b_{i2}),$$

where $n = 46$.

The maximum likelihood estimator is

$$(\hat{\alpha}_1, \hat{\alpha}_2) = \arg \max_{\alpha_1, \alpha_2} L(\alpha_1, \alpha_2).$$

Because maximizing the likelihood directly can involve very small numbers, work with the log-likelihood

$$\ell(\alpha_1, \alpha_2) = \sum_{i=1}^n [N_i(\alpha_1 b_{i1} + \alpha_2 b_{i2}) - \exp(\alpha_1 b_{i1} + \alpha_2 b_{i2}) - \log(N_i!)] .$$

2(a) Derivatives and Newton–Raphson

Let

$$\eta_i = \alpha_1 b_{i1} + \alpha_2 b_{i2}, \quad \lambda_i = e^{\eta_i}.$$

- i. Derive the score vector

$$\nabla \ell(\alpha_1, \alpha_2) = \begin{pmatrix} \partial \ell / \partial \alpha_1 \\ \partial \ell / \partial \alpha_2 \end{pmatrix}.$$

- ii. Derive the Hessian matrix

$$H(\alpha_1, \alpha_2) = \begin{pmatrix} \partial^2 \ell / \partial \alpha_1^2 & \partial^2 \ell / \partial \alpha_1 \partial \alpha_2 \\ \partial^2 \ell / \partial \alpha_2 \partial \alpha_1 & \partial^2 \ell / \partial \alpha_2^2 \end{pmatrix}.$$

- iii. Write the Newton–Raphson update in matrix form. If

$$\boldsymbol{\alpha}^{(t)} = \begin{pmatrix} \alpha_1^{(t)} \\ \alpha_2^{(t)} \end{pmatrix},$$

show that the update can be written as

$$\boldsymbol{\alpha}^{(t+1)} = \boldsymbol{\alpha}^{(t)} - H(\boldsymbol{\alpha}^{(t)})^{-1} \nabla \ell(\boldsymbol{\alpha}^{(t)}).$$

- iv. Briefly explain why the Hessian is expected to be negative semidefinite (and typically negative definite when the two predictors provide sufficiently distinct information). What does this imply about the shape of the log-likelihood?

2(b) Implement Newton–Raphson in R

Implement Newton–Raphson without using a built-in Poisson regression function such as `glm()` to obtain the estimates.

Use the starting value

$$\boldsymbol{\alpha}^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

At each iteration, calculate and store:

$$t, \quad \alpha_1^{(t)}, \quad \alpha_2^{(t)}, \quad \ell(\boldsymbol{\alpha}^{(t)}), \quad \|\nabla \ell(\boldsymbol{\alpha}^{(t)})\|_2.$$

Stop when

$$\|\boldsymbol{\alpha}^{(t+1)} - \boldsymbol{\alpha}^{(t)}\|_2 < 10^{-6},$$

or after 100 iterations, whichever occurs first.

Report:

- the final MLEs $\hat{\alpha}_1$ and $\hat{\alpha}_2$;
- the number of iterations required;
- the final value of the log-likelihood;
- a table of the first few and last few iterations.

As a numerical check, verify that the gradient at your reported estimate is very close to zero.

2(c) Steepest Ascent with Step-Halving

Apply the method of steepest ascent to maximize $\ell(\boldsymbol{\alpha})$.

For this homework, define one steepest-ascent iteration as

$$\boldsymbol{\alpha}^{(t+1)} = \boldsymbol{\alpha}^{(t)} + s_t \nabla \ell(\boldsymbol{\alpha}^{(t)}),$$

where $s_t > 0$ is selected by step-halving.

Start with

$$s_t = 1.$$

If

$$\ell(\boldsymbol{\alpha}^{(t)} + s_t \nabla \ell(\boldsymbol{\alpha}^{(t)})) \leq \ell(\boldsymbol{\alpha}^{(t)}),$$

replace s_t by $s_t/2$ and repeat until the log-likelihood increases.

For numerical stability, impose a maximum of 50 halvings during any iteration. Use the same starting value and convergence criterion as in part (b).

Report the final estimate, number of iterations, final log-likelihood, and the number of step-halving operations. Include enough information to assess whether the algorithm is behaving sensibly.

2(d) Gradient Ascent with a Fixed Step Size

Now implement gradient ascent using a fixed step size:

$$\boldsymbol{\alpha}^{(t+1)} = \boldsymbol{\alpha}^{(t)} + \gamma \nabla \ell(\boldsymbol{\alpha}^{(t)}).$$

Choose a constant γ that gives stable convergence. Explain briefly how you selected γ . You may experiment with several values, but report the value used for your final results.

Use the same starting value and convergence criterion as in part (b). If the algorithm fails to converge within 1000 iterations, report this explicitly rather than changing the stopping rule.

2(e) Compare the Three Optimization Methods

Compare Newton–Raphson, steepest ascent with step-halving, and fixed-step gradient ascent.

Create a table containing, at minimum,

Method	Iterations	Log-likelihood	$\hat{\alpha}_1$	$\hat{\alpha}_2$
Newton–Raphson				
Steepest ascent				
Gradient ascent				

Then create a plot of the log-likelihood versus iteration number for the three methods. Because the algorithms may require very different numbers of iterations, use an appropriate plotting scale or clearly explain any restriction on the displayed iterations.

In a short discussion, address:

- Which method converged fastest?
- Which method required the most function/gradient evaluations or step-size adjustments?
- Did all three methods produce essentially the same MLE?
- Why might Newton–Raphson converge much faster near the maximum?

Submission Requirements

Submit one PDF containing your answers and one R script containing the code used to produce your results.

Your solution should:

- clearly distinguish mathematical derivations from R output;
- include labeled and readable figures;
- report numerical results with reasonable precision;
- include enough code that the results can be reproduced;
- include short interpretations rather than reporting numerical answers without explanation.